PSO – statistické tabulky

kvantil x_p: $p(X < x_p) = p$ · oboustranně $\alpha \rightarrow p = 1 - \alpha/2$ · jednostranně $\rightarrow p = 1 - \alpha$

$\Phi(u) = p(U < u)$, $U \sim No(0, 1)$ · $\Phi(-u) = 1 - \Phi(u)$ · hodnoty bez „0"

u	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	,5000	,5040	,5080	,5120	,5160	,5199	,5239	,5279	,5319	,5359
0,1	,5398	,5438	,5478	,5517	,5557	,5596	,5636	,5675	,5714	,5753
0,2	,5793	,5832	,5871	,5910	,5948	,5987	,6026	,6064	,6103	,6141
0,3	,6179	,6217	,6255	,6293	,6331	,6368	,6406	,6443	,6480	,6517
0,4	,6554	,6591	,6628	,6664	,6700	,6736	,6772	,6808	,6844	,6879
0,5	,6915	,6950	,6985	,7019	,7054	,7088	,7123	,7157	,7190	,7224
0,6	,7257	,7291	,7324	,7357	,7389	,7422	,7454	,7486	,7517	,7549
0,7	,7580	,7611	,7642	,7673	,7704	,7734	,7764	,7794	,7823	,7852
0,8	,7881	,7910	,7939	,7967	,7995	,8023	,8051	,8078	,8106	,8133
0,9	,8159	,8186	,8212	,8238	,8264	,8289	,8315	,8340	,8365	,8389
1,0	,8413	,8438	,8461	,8485	,8508	,8531	,8554	,8577	,8599	,8621
1,1	,8643	,8665	,8686	,8708	,8729	,8749	,8770	,8790	,8810	,8830
1,2	,8849	,8869	,8888	,8907	,8925	,8944	,8962	,8980	,8997	,9015
1,3	,9032	,9049	,9066	,9082	,9099	,9115	,9131	,9147	,9162	,9177
1,4	,9192	,9207	,9222	,9236	,9251	,9265	,9279	,9292	,9306	,9319
1,5	,9332	,9345	,9357	,9370	,9382	,9394	,9406	,9418	,9429	,9441
1,6	,9452	,9463	,9474	,9484	,9495	,9505	,9515	,9525	,9535	,9545
1,7	,9554	,9564	,9573	,9582	,9591	,9599	,9608	,9616	,9625	,9633
1,8	,9641	,9649	,9656	,9664	,9671	,9678	,9686	,9693	,9699	,9706
1,9	,9713	,9719	,9726	,9732	,9738	,9744	,9750	,9756	,9761	,9767
2,0	,9772	,9778	,9783	,9788	,9793	,9798	,9803	,9808	,9812	,9817
2,1	,9821	,9826	,9830	,9834	,9838	,9842	,9846	,9850	,9854	,9857
2,2	,9861	,9864	,9868	,9871	,9875	,9878	,9881	,9884	,9887	,9890
2,3	,9893	,9896	,9898	,9901	,9904	,9906	,9909	,9911	,9913	,9916
2,4	,9918	,9920	,9922	,9925	,9927	,9929	,9931	,9932	,9934	,9936
2,5	,9938	,9940	,9941	,9943	,9945	,9946	,9948	,9949	,9951	,9952
2,6	,9953	,9955	,9956	,9957	,9959	,9960	,9961	,9962	,9963	,9964
2,7	,9965	,9966	,9967	,9968	,9969	,9970	,9971	,9972	,9973	,9974
2,8	,9974	,9975	,9976	,9977	,9977	,9978	,9979	,9979	,9980	,9981
2,9	,9981	,9982	,9982	,9983	,9984	,9984	,9985	,9985	,9986	,9986
3,0	,9987	,9987	,9987	,9988	,9988	,9989	,9989	,9989	,9990	,9990
3,1	,9990	,9991	,9991	,9991	,9992	,9992	,9992	,9992	,9993	,9993
3,2	,9993	,9993	,9994	,9994	,9994	,9994	,9994	,9995	,9995	,9995
3,3	,9995	,9995	,9995	,9996	,9996	,9996	,9996	,9996	,9996	,9997
3,4	,9997	,9997	,9997	,9997	,9997	,9997	,9997	,9997	,9997	,9998

Kvantily: $u_{1-p} = -u_p$

p	0,8	0,85	0,875	0,9	0,95	0,975	0,98	0,99	0,995	0,999
u_p	0,8416	1,0364	1,1503	1,2816	1,6449	1,9600	2,0537	2,3263	2,5758	3,0902

Studentovo t(ν): $t_p(\nu)$

ν	0,9	0,95	0,975	0,99	0,995
1	3,078	6,314	12,706	31,821	63,657
2	1,886	2,920	4,303	6,965	9,925
3	1,638	2,353	3,182	4,541	5,841
4	1,533	2,132	2,776	3,747	4,604
5	1,476	2,015	2,571	3,365	4,032
6	1,440	1,943	2,447	3,143	3,707
7	1,415	1,895	2,365	2,998	3,499
8	1,397	1,860	2,306	2,896	3,355
9	1,383	1,833	2,262	2,821	3,250
10	1,372	1,812	2,228	2,764	3,169
11	1,363	1,796	2,201	2,718	3,106
12	1,356	1,782	2,179	2,681	3,055
13	1,350	1,771	2,160	2,650	3,012
14	1,345	1,761	2,145	2,624	2,977
15	1,341	1,753	2,131	2,602	2,947
16	1,337	1,746	2,120	2,583	2,921
17	1,333	1,740	2,110	2,567	2,898
18	1,330	1,734	2,101	2,552	2,878
19	1,328	1,729	2,093	2,539	2,861
20	1,325	1,725	2,086	2,528	2,845
21	1,323	1,721	2,080	2,518	2,831
22	1,321	1,717	2,074	2,508	2,819
23	1,319	1,714	2,069	2,500	2,807
24	1,318	1,711	2,064	2,492	2,797
25	1,316	1,708	2,060	2,485	2,787
26	1,315	1,706	2,056	2,479	2,779
27	1,314	1,703	2,052	2,473	2,771
28	1,313	1,701	2,048	2,467	2,763
29	1,311	1,699	2,045	2,462	2,756
30	1,310	1,697	2,042	2,457	2,750
40	1,303	1,684	2,021	2,423	2,704
50	1,299	1,676	2,009	2,403	2,678
60	1,296	1,671	2,000	2,390	2,660
80	1,292	1,664	1,990	2,374	2,639
100	1,290	1,660	1,984	2,364	2,626
120	1,289	1,658	1,980	2,358	2,617
150	1,287	1,655	1,976	2,351	2,609
200	1,286	1,653	1,972	2,345	2,601
∞	1,282	1,645	1,960	2,326	2,576

$\chi^2(\nu)$: $\chi^2_p(\nu)$

ν	0,005	0,01	0,025	0,05	0,95	0,975	0,99	0,995
1	0,0000	0,0002	0,0010	0,0039	3,841	5,024	6,635	7,879
2	0,0100	0,0201	0,0506	0,103	5,991	7,378	9,210	10,60
3	0,0717	0,115	0,216	0,352	7,815	9,348	11,34	12,84
4	0,207	0,297	0,484	0,711	9,488	11,14	13,28	14,86
5	0,412	0,554	0,831	1,145	11,07	12,83	15,09	16,75
6	0,676	0,872	1,237	1,635	12,59	14,45	16,81	18,55
7	0,989	1,239	1,690	2,167	14,07	16,01	18,48	20,28
8	1,344	1,646	2,180	2,733	15,51	17,53	20,09	21,95
9	1,735	2,088	2,700	3,325	16,92	19,02	21,67	23,59
10	2,156	2,558	3,247	3,940	18,31	20,48	23,21	25,19
11	2,603	3,053	3,816	4,575	19,68	21,92	24,72	26,76
12	3,074	3,571	4,404	5,226	21,03	23,34	26,22	28,30
13	3,565	4,107	5,009	5,892	22,36	24,74	27,69	29,82
14	4,075	4,660	5,629	6,571	23,68	26,12	29,14	31,32
15	4,601	5,229	6,262	7,261	25,00	27,49	30,58	32,80
16	5,142	5,812	6,908	7,962	26,30	28,85	32,00	34,27
17	5,697	6,408	7,564	8,672	27,59	30,19	33,41	35,72
18	6,265	7,015	8,231	9,390	28,87	31,53	34,81	37,16
19	6,844	7,633	8,907	10,12	30,14	32,85	36,19	38,58
20	7,434	8,260	9,591	10,85	31,41	34,17	37,57	40,00
21	8,034	8,897	10,28	11,59	32,67	35,48	38,93	41,40
22	8,643	9,542	10,98	12,34	33,92	36,78	40,29	42,80
23	9,260	10,20	11,69	13,09	35,17	38,08	41,64	44,18
24	9,886	10,86	12,40	13,85	36,42	39,36	42,98	45,56
25	10,52	11,52	13,12	14,61	37,65	40,65	44,31	46,93
26	11,16	12,20	13,84	15,38	38,89	41,92	45,64	48,29
27	11,81	12,88	14,57	16,15	40,11	43,19	46,96	49,64
28	12,46	13,56	15,31	16,93	41,34	44,46	48,28	50,99
29	13,12	14,26	16,05	17,71	42,56	45,72	49,59	52,34
30	13,79	14,95	16,79	18,49	43,77	46,98	50,89	53,67
40	20,71	22,16	24,43	26,51	55,76	59,34	63,69	66,77
50	27,99	29,71	32,36	34,76	67,50	71,42	76,15	79,49
60	35,53	37,48	40,48	43,19	79,08	83,30	88,38	91,95
80	51,17	53,54	57,15	60,39	101,9	106,6	112,3	116,3
100	67,33	70,06	74,22	77,93	124,3	129,6	135,8	140,2
120	83,85	86,92	91,57	95,70	146,6	152,2	159,0	163,6

Pro $\nu > 100$: $\chi^2_p(\nu) \approx \frac{1}{2}(u_p + \sqrt{(2\nu - 1)})^2$.

$F(\nu_1, \nu_2)$: $F_{0,975}$ · oboustranný F-test s $\alpha = 0{,}05$ · ν_1 = stupně volnosti čitatele (větší rozptyl)

$\nu_2 \downarrow \nu_1 \rightarrow$	1	2	3	4	5	6	7	8	9	10	12	15	20	30	60	∞
1	647,8	799,5	864,2	899,6	921,8	937,1	948,2	956,7	963,3	968,6	976,7	984,9	993,1	1001,4	1009,8	1018,3
2	38,51	39,00	39,17	39,25	39,30	39,33	39,36	39,37	39,39	39,40	39,41	39,43	39,45	39,46	39,48	39,50
3	17,44	16,04	15,44	15,10	14,88	14,73	14,62	14,54	14,47	14,42	14,34	14,25	14,17	14,08	13,99	13,90
4	12,22	10,65	9,979	9,605	9,364	9,197	9,074	8,980	8,905	8,844	8,751	8,657	8,560	8,461	8,360	8,257
5	10,01	8,434	7,764	7,388	7,146	6,978	6,853	6,757	6,681	6,619	6,525	6,428	6,329	6,227	6,123	6,015
6	8,813	7,260	6,599	6,227	5,988	5,820	5,695	5,600	5,523	5,461	5,366	5,269	5,168	5,065	4,959	4,849
7	8,073	6,542	5,890	5,523	5,285	5,119	4,995	4,899	4,823	4,761	4,666	4,568	4,467	4,362	4,254	4,142
8	7,571	6,059	5,416	5,053	4,817	4,652	4,529	4,433	4,357	4,295	4,200	4,101	3,999	3,894	3,784	3,670
9	7,209	5,715	5,078	4,718	4,484	4,320	4,197	4,102	4,026	3,964	3,868	3,769	3,667	3,560	3,449	3,333
10	6,937	5,456	4,826	4,468	4,236	4,072	3,950	3,855	3,779	3,717	3,621	3,522	3,419	3,311	3,198	3,080
11	6,724	5,256	4,630	4,275	4,044	3,881	3,759	3,664	3,588	3,526	3,430	3,330	3,226	3,118	3,004	2,883
12	6,554	5,096	4,474	4,121	3,891	3,728	3,607	3,512	3,436	3,374	3,277	3,177	3,073	2,963	2,848	2,725
13	6,414	4,965	4,347	3,996	3,767	3,604	3,483	3,388	3,312	3,250	3,153	3,053	2,948	2,837	2,720	2,595
14	6,298	4,857	4,242	3,892	3,663	3,501	3,380	3,285	3,209	3,147	3,050	2,949	2,844	2,732	2,614	2,487
15	6,200	4,765	4,153	3,804	3,576	3,415	3,293	3,199	3,123	3,060	2,963	2,862	2,756	2,644	2,524	2,395
16	6,115	4,687	4,077	3,729	3,502	3,341	3,219	3,125	3,049	2,986	2,889	2,788	2,681	2,568	2,447	2,316
17	6,042	4,619	4,011	3,665	3,438	3,277	3,156	3,061	2,985	2,922	2,825	2,723	2,616	2,502	2,380	2,247
18	5,978	4,560	3,954	3,608	3,382	3,221	3,100	3,005	2,929	2,866	2,769	2,667	2,559	2,445	2,321	2,187
19	5,922	4,508	3,903	3,559	3,333	3,172	3,051	2,956	2,880	2,817	2,720	2,617	2,509	2,394	2,270	2,133
20	5,871	4,461	3,859	3,515	3,289	3,128	3,007	2,913	2,837	2,774	2,676	2,573	2,464	2,349	2,223	2,085
21	5,827	4,420	3,819	3,475	3,250	3,090	2,969	2,874	2,798	2,735	2,637	2,534	2,425	2,308	2,182	2,042
22	5,786	4,383	3,783	3,440	3,215	3,055	2,934	2,839	2,763	2,700	2,602	2,498	2,389	2,272	2,145	2,003
23	5,750	4,349	3,750	3,408	3,183	3,023	2,902	2,808	2,731	2,668	2,570	2,466	2,357	2,239	2,111	1,968
24	5,717	4,319	3,721	3,379	3,155	2,995	2,874	2,779	2,703	2,640	2,541	2,437	2,327	2,209	2,080	1,935
25	5,686	4,291	3,694	3,353	3,129	2,969	2,848	2,753	2,677	2,613	2,515	2,411	2,300	2,182	2,052	1,906
26	5,659	4,265	3,670	3,329	3,105	2,945	2,824	2,729	2,653	2,590	2,491	2,387	2,276	2,157	2,026	1,878
27	5,633	4,242	3,647	3,307	3,083	2,923	2,802	2,707	2,631	2,568	2,469	2,364	2,253	2,133	2,002	1,853
28	5,610	4,221	3,626	3,286	3,063	2,903	2,782	2,687	2,611	2,547	2,448	2,344	2,232	2,112	1,980	1,829
29	5,588	4,201	3,607	3,267	3,044	2,884	2,763	2,669	2,592	2,529	2,430	2,325	2,213	2,092	1,959	1,807
30	5,568	4,182	3,589	3,250	3,026	2,867	2,746	2,651	2,575	2,511	2,412	2,307	2,195	2,074	1,940	1,787
40	5,424	4,051	3,463	3,126	2,904	2,744	2,624	2,529	2,452	2,388	2,288	2,182	2,068	1,943	1,803	1,637
60	5,286	3,925	3,343	3,008	2,786	2,627	2,507	2,412	2,334	2,270	2,169	2,061	1,944	1,815	1,667	1,482
120	5,152	3,805	3,227	2,894	2,674	2,515	2,395	2,299	2,222	2,157	2,055	1,945	1,825	1,690	1,530	1,310
∞	5,024	3,689	3,116	2,786	2,567	2,408	2,288	2,192	2,114	2,048	1,945	1,833	1,708	1,566	1,388	1,001

$F(\nu_1, \nu_2)$: $F_{0,95}$ · jednostranný F-test s $\alpha = 0{,}05$ · $F_\alpha(\nu_1, \nu_2) = 1 / F_{1-\alpha}(\nu_2, \nu_1)$

$\nu_2 \downarrow \nu_1 \rightarrow$	1	2	3	4	5	6	7	8	9	10	12	15	20	30	60	∞
1	161,4	199,5	215,7	224,6	230,2	234,0	236,8	238,9	240,5	241,9	243,9	245,9	248,0	250,1	252,2	254,3
2	18,51	19,00	19,16	19,25	19,30	19,33	19,35	19,37	19,38	19,40	19,41	19,43	19,45	19,46	19,48	19,50
3	10,13	9,552	9,277	9,117	9,013	8,941	8,887	8,845	8,812	8,786	8,745	8,703	8,660	8,617	8,572	8,526
4	7,709	6,944	6,591	6,388	6,256	6,163	6,094	6,041	5,999	5,964	5,912	5,858	5,803	5,746	5,688	5,628
5	6,608	5,786	5,409	5,192	5,050	4,950	4,876	4,818	4,772	4,735	4,678	4,619	4,558	4,496	4,431	4,365
6	5,987	5,143	4,757	4,534	4,387	4,284	4,207	4,147	4,099	4,060	4,000	3,938	3,874	3,808	3,740	3,669
7	5,591	4,737	4,347	4,120	3,972	3,866	3,787	3,726	3,677	3,637	3,575	3,511	3,445	3,376	3,304	3,230
8	5,318	4,459	4,066	3,838	3,687	3,581	3,500	3,438	3,388	3,347	3,284	3,218	3,150	3,079	3,005	2,928
9	5,117	4,256	3,863	3,633	3,482	3,374	3,293	3,230	3,179	3,137	3,073	3,006	2,936	2,864	2,787	2,707
10	4,965	4,103	3,708	3,478	3,326	3,217	3,135	3,072	3,020	2,978	2,913	2,845	2,774	2,700	2,621	2,538
11	4,844	3,982	3,587	3,357	3,204	3,095	3,012	2,948	2,896	2,854	2,788	2,719	2,646	2,570	2,490	2,404
12	4,747	3,885	3,490	3,259	3,106	2,996	2,913	2,849	2,796	2,753	2,687	2,617	2,544	2,466	2,384	2,296
13	4,667	3,806	3,411	3,179	3,025	2,915	2,832	2,767	2,714	2,671	2,604	2,533	2,459	2,380	2,297	2,206
14	4,600	3,739	3,344	3,112	2,958	2,848	2,764	2,699	2,646	2,602	2,534	2,463	2,388	2,308	2,223	2,131
15	4,543	3,682	3,287	3,056	2,901	2,790	2,707	2,641	2,588	2,544	2,475	2,403	2,328	2,247	2,160	2,066
16	4,494	3,634	3,239	3,007	2,852	2,741	2,657	2,591	2,538	2,494	2,425	2,352	2,276	2,194	2,106	2,010
17	4,451	3,592	3,197	2,965	2,810	2,699	2,614	2,548	2,494	2,450	2,381	2,308	2,230	2,148	2,058	1,960
18	4,414	3,555	3,160	2,928	2,773	2,661	2,577	2,510	2,456	2,412	2,342	2,269	2,191	2,107	2,017	1,917
19	4,381	3,522	3,127	2,895	2,740	2,628	2,544	2,477	2,423	2,378	2,308	2,234	2,155	2,071	1,980	1,878
20	4,351	3,493	3,098	2,866	2,711	2,599	2,514	2,447	2,393	2,348	2,278	2,203	2,124	2,039	1,946	1,843
21	4,325	3,467	3,072	2,840	2,685	2,573	2,488	2,420	2,366	2,321	2,250	2,176	2,096	2,010	1,916	1,812
22	4,301	3,443	3,049	2,817	2,661	2,549	2,464	2,397	2,342	2,297	2,226	2,151	2,071	1,984	1,889	1,783
23	4,279	3,422	3,028	2,796	2,640	2,528	2,442	2,375	2,320	2,275	2,204	2,128	2,048	1,961	1,865	1,757
24	4,260	3,403	3,009	2,776	2,621	2,508	2,423	2,355	2,300	2,255	2,183	2,108	2,027	1,939	1,842	1,733
25	4,242	3,385	2,991	2,759	2,603	2,490	2,405	2,337	2,282	2,236	2,165	2,089	2,007	1,919	1,822	1,711
26	4,225	3,369	2,975	2,743	2,587	2,474	2,388	2,321	2,265	2,220	2,148	2,072	1,990	1,901	1,803	1,691
27	4,210	3,354	2,960	2,728	2,572	2,459	2,373	2,305	2,250	2,204	2,132	2,056	1,974	1,884	1,785	1,672
28	4,196	3,340	2,947	2,714	2,558	2,445	2,359	2,291	2,236	2,190	2,118	2,041	1,959	1,869	1,769	1,654
29	4,183	3,328	2,934	2,701	2,545	2,432	2,346	2,278	2,223	2,177	2,104	2,027	1,945	1,854	1,754	1,638
30	4,171	3,316	2,922	2,690	2,534	2,421	2,334	2,266	2,211	2,165	2,092	2,015	1,932	1,841	1,740	1,622
40	4,085	3,232	2,839	2,606	2,449	2,336	2,249	2,180	2,124	2,077	2,003	1,924	1,839	1,744	1,637	1,509
60	4,001	3,150	2,758	2,525	2,368	2,254	2,167	2,097	2,040	1,993	1,917	1,836	1,748	1,649	1,534	1,389
120	3,920	3,072	2,680	2,447	2,290	2,175	2,087	2,016	1,959	1,910	1,834	1,750	1,659	1,554	1,429	1,254
∞	3,841	2,996	2,605	2,372	2,214	2,099	2,010	1,938	1,880	1,831	1,752	1,666	1,571	1,459	1,318	1,001

Kolmogorovův–Smirnovův test: $D_n(\alpha)$ · zamítám, když $D = \max|F_n - F_0| \geq D_n(\alpha)$; $n > 40$: $D_n(0{,}05) \approx 1{,}36/\sqrt{n}$, $D_n(0{,}01) \approx 1{,}63/\sqrt{n}$

n	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
$\alpha = 0{,}05$	0,975	0,842	0,708	0,624	0,563	0,519	0,483	0,454	0,430	0,409	0,391	0,375	0,361	0,349	0,338	0,327	0,318	0,309	0,301	0,294
$\alpha = 0{,}01$	0,995	0,929	0,829	0,734	0,669	0,617	0,576	0,542	0,513	0,489	0,468	0,449	0,432	0,418	0,404	0,392	0,381	0,371	0,361	0,352
n	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
$\alpha = 0{,}05$	0,287	0,281	0,275	0,269	0,264	0,259	0,254	0,250	0,246	0,242	0,238	0,234	0,231	0,227	0,224	0,221	0,218	0,215	0,213	0,210
$\alpha = 0{,}01$	0,344	0,337	0,330	0,323	0,317	0,311	0,305	0,300	0,295	0,290	0,285	0,281	0,277	0,273	0,269	0,265	0,262	0,258	0,255	0,252